

Dual-Track Longitudinal Modeling of Diabetes Risk: Horizon-Specific Screening and Intervention-Safe Scoring

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Abstract—Effective diabetes prevention relies on identifying metabolic risk during the pre-diagnostic window, yet existing prediction tools typically fix a single forecast horizon and often assume a single model can serve both population screening and patient counseling. This work introduced an end-to-end longitudinal diabetes-risk modeling pipeline developed from 12 years of annual health-check data from 6,892 patients at a tertiary hospital in Thailand (2005–2016). We framed the task as a rolling patient-year classification, explicitly varying the prediction horizon (1 to 5 years) and historical input windows (1, 3, or 5 years) across biostatistical, tree-based, and survival model families. Our benchmarks revealed that no single model family dominated: tree-based models led at short horizons and biostatistical models at longer ones. Importantly, we demonstrated that ranking future risk and patient-facing what-if scoring were distinct operational tasks. To address this, we proposed a dual-track framework: a horizon-specific pure-prediction track for passive screening, and an “intervention-safe” monotonic track designed for patient-facing what-if simulations. The intervention-safe models preserved clinically plausible score directions under tested lifestyle-improvement scenarios while matching or exceeding pure-prediction performance at horizons up to four years. Across horizons the leading models reached ROC-AUC 0.80–0.83 with rank-1 PR-AUC lifts of 3.2–5.4× over the prevalence baseline, and patient-clustered bootstrap confidence intervals showed the tree and biostatistical leaders to be statistically indistinguishable at every horizon. This dual-track architecture bridged the gap between passive risk stratification and actionable clinical decision support. Code is publicly available at <https://www.github.com/pitikorn32/dual-track-longitudinal-diabetes-risk>.

Index Terms—diabetes risk prediction, longitudinal health-check data, rolling patient-year modeling, gradient boosting, generalized estimating equations, survival analysis, monotonic constraints, intervention-safe modeling, PR-AUC

I. INTRODUCTION

Diabetes mellitus imposes a growing burden on health systems worldwide, and its complications hinge on whether elevated metabolic risk is detected early enough for preventive action [1]. Because lifestyle changes can prevent or delay

progression to type 2 diabetes in high-risk individuals [2], [3], the pre-diagnostic period, when a patient is drifting toward glycemic instability but not yet diagnosed, is the most valuable window for intervention. Annual health-check records suit this setting because they capture how risk evolves over time rather than a single cross-sectional snapshot.

Two methodological gaps limit the practical use of existing tools. The first concerns *temporal framing*. Many studies evaluate one fixed horizon, one baseline cohort, or one feature snapshot [4], [5], whereas preventive care requires repeated decisions: a clinician may ask whether a patient becomes at risk in one year, three years, or five years, and the answer depends on how much prior history is available. This defines two temporal dimensions, the prediction horizon N and the history window M , that are rarely compared systematically within one design.

The second gap concerns *intervention use*. Models optimized for discrimination can learn associations that are predictive but may be causally misleading. Patients who feel unwell may reduce sugar intake or lose weight before diagnosis, so an unconstrained model can link a favorable change with higher predicted risk [6], [7]. Such behavior is unacceptable when the same model is used for “what-if” counseling.

Accordingly, we addressed both gaps with a unified pipeline on rolling patient-year data (Fig. 1). To our knowledge, this is the first study to jointly sweep prediction horizon and history window across biostatistical, tree-based, and survival families on one shared cohort while adding an explicit intervention-safe track. Our contributions were:

- a longitudinal evaluation framework that varied horizon N and history M on a single *shared cohort*, enabling a fair comparison of biostatistical, tree-based, and survival families under one protocol;
- an empirical finding, supported by patient-clustered bootstrap confidence intervals, that no family dominated: trees led or stayed near the top at short horizons and statistical models led at long horizons, yet the per-horizon leaders were statistically indistinguishable, and longer history most helped the direct-prediction finalists;

End-to-End Dual-Track Longitudinal Modeling of Diabetes Risk Pipeline

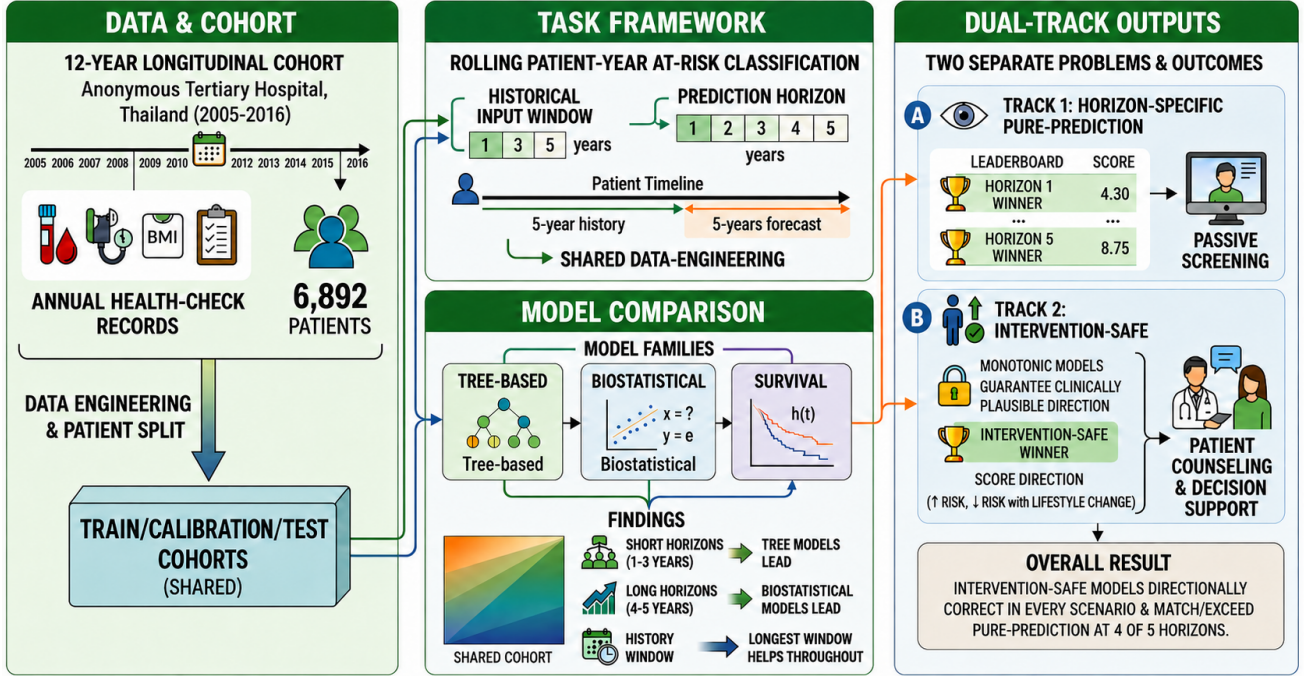


Fig. 1. End-to-end study pipeline. A 12-year longitudinal cohort is transformed into rolling patient-year tables with configurable history windows and prediction horizons, evaluated across tree-based, biostatistical, and survival model families, and summarized into two tracks: a horizon-specific pure-prediction leaderboard and an intervention-safe scoring benchmark.

- a *dual-track* formulation that separated pure-prediction screening from intervention-safe scoring, with a scenario-based directional-safety check; monotonic models preserved clinically plausible score direction while matching or exceeding the pure-prediction winners at four of five horizons.

II. RELATED WORK

Diabetes-risk modeling spans three traditions. *Biostatistical* models such as logistic regression [8] remain attractive for interpretability; generalized estimating equations (GEE) provide a population-averaged fit under within-patient correlation [9], while generalized linear mixed models (GLMMs) add subject-specific random effects [10]. *Tree ensembles* such as XGBoost [11], CatBoost [12], and LightGBM [13] capture nonlinear and threshold effects in tabular clinical data but often need post-hoc calibration [14], [15]. *Survival* methods, from the Cox model [16] to landmarking [17] and recent two-stage dynamic prediction [18], treat onset timing as the outcome.

A second axis is static versus longitudinal prediction. Static baselines discard temporal structure, whereas longitudinal formulations exploit repeated measurements and trends [19], [20]. Missingness in such data is often informative and must be explicitly represented rather than silently imputed [21]. Despite this, few studies jointly sweep prediction horizon and lookback history within a unified pipeline [5].

A third axis addresses intervention safety through monotonic constraints [22] and interpretable additive models, such

as explainable boosting machines [23], [24], which force selected features to act only in clinically plausible directions. We brought these strands together by evaluating a shared-cohort horizon/history sweep across multiple model families, while simultaneously maintaining an explicit, intervention-safe evaluation track.

Recurrent and Transformer-based sequence models have advanced longitudinal EHR prediction [25], [26], but annual health-check panels such as ours are smaller and less densely sampled, so we leave deep sequence models to future work. Relative to fixed-form instruments such as FINDRISC [27] and the Thai-population risk score, our shared-cohort ROC-AUC of 0.80–0.83 is numerically higher than the latter’s reported AUC of 0.74 [28]; these cross-study values are descriptive rather than head-to-head comparisons because the cohorts, horizons, predictors, and outcomes differ. Unlike such fixed-form scores, our framework controls horizon and history and includes an intervention-safe track.

III. DATA AND PROBLEM FORMULATION

A. Dataset

We used annual health check-up records from a tertiary hospital in Thailand (2005–2016), comprising 6,892 patients in a wide-format table with year-suffixed measurements. Variables spanned demographic and lifestyle fields (sex, family history, smoking, alcohol, exercise, physical activity, diet, sleep) and annual clinical measurements (fasting blood sugar [FBS], body mass index [BMI], pulse, systolic and diastolic blood pressure,

waist). Missingness was year-dependent, averaging 33–45% per variable and falling from 51–69% in 2005–2007 to about 3% by 2016; to avoid bias toward high-attendance patients, it was encoded explicitly rather than imputed.

B. Outcome and rolling formulation

The target was a binary `AtRisk` status that merged pre-diabetes and diabetes into one positive class. In the source cohort, `DM_status` was derived from the maximum fasting blood sugar recorded for each patient from the beginning of the study period (2005) through each year (`MAX_FBS_up_to_year`): non-DM ≤ 100 mg/dL, pre-DM > 100 to ≤ 125 mg/dL, and DM > 125 mg/dL (equivalent to ≥ 126 mg/dL for integer measurements). This target captured the first transition out of a metabolically normal state. The wide table was reshaped to one row per patient-year. For a patient i observed at source year T , the label at horizon N was

$$Y_{i,T}^{(N)} = \text{AtRisk}_{i,T+N}, \quad N \in \{1, 2, 3, 4, 5\}. \quad (1)$$

The history window $M \in \{1, 3, 5\}$ controlled how many prior years were summarized into rolling features (latest, mean, min, max, std, range, and Ordinary Least Squares (OLS) slope per clinical variable), so the $N \times M$ grid yielded 15 modeling tables. We modeled *first onset*: a row (i, T) was eligible only if $\text{AtRisk}_{i,T} = 0$, the target $Y_{i,T}^{(N)}$ was observed, and T preceded the patient's first at-risk year. Patients already at risk in 2005 contributed no eligible rows and were excluded.

C. Splitting and leakage control

To ensure consistency, a canonical patient-grouped split (60/20/20 train/calibration/test, fixed seed) was generated once and reused. For model families that did not require a held-out calibration set, these samples were folded back into the training set for an effective 80/20 split. Data leakage was prevented by design because labels were derived only from year $T+N$, predictors were restricted to years $\leq T$, post-onset rows were censored, and patient identities never crossed split boundaries.

IV. METHODS

A. Model families

Biostatistical: The statistical branch used interpretable models for repeated binary outcomes. Penalized logistic regression served as the fixed-effect baseline [8], using the logistic specification

$$\text{logit}\{P(Y_{iT} = 1)\} = \beta_0 + \mathbf{x}_{i,T}^\top \boldsymbol{\beta}, \quad (2)$$

fitted by minimizing the mean binary cross-entropy with an L_2 penalty ($\lambda = 0.01$, intercept unpenalized), with cluster-robust covariance estimation to account for repeated measurements within patients. GEE used the same marginal mean model but represented within-patient dependence through an exchangeable working correlation [9]:

$$\begin{aligned} \text{logit}(\mu_{i,T}) &= \beta_0 + \mathbf{x}_{i,T}^\top \boldsymbol{\beta}, \\ \text{Corr}(Y_{i,T}, Y_{i,T'}) &= \alpha \text{ for } T \neq T'. \end{aligned} \quad (3)$$

where $\mu_{i,T} = \mathbb{E}[Y_{i,T} \mid \mathbf{x}_{i,T}]$. A GLMM comparator added a patient-level random intercept $b_i \sim \mathcal{N}(0, \sigma_b^2)$ to the same logit mean [10], fit by Laplace-approximation maximum likelihood (GPBoost). Because the random intercept could not be estimated for unseen patients under a patient-grouped split, its held-out predictions collapsed to the marginal (Pinheiro–Bates) mean and trailed GEE and logistic at every horizon, so the GLMM was retained only as a documented benchmark [29].

Tree-based: XGBoost [11] and CatBoost [12] were the retained learners; LightGBM [13] trailed both at every grid cell. Class imbalance was handled by positive-class weighting,

$$w_+ = \frac{n_{\text{neg}}}{n_{\text{pos}}}, \quad (4)$$

where n_{pos} and n_{neg} are the positive- and negative-class counts in the training rows. A hybrid branch augmented these learners with empirical-Bayes-shrunk longitudinal slope features, where patient-specific OLS slopes were shrunk toward the population slope [30].

Survival: A rolling landmark Cox model [16], [17] and a two-stage model [18] both operated on the same rolling rows. The two-stage variant first predicted near-future covariates and then passed them to the Cox stage. Survival estimates were converted to fixed-horizon event probabilities:

$$h_i(t) = h_0(t) \exp(\mathbf{x}_{i,T}^\top \boldsymbol{\beta}), \quad \hat{p}_i(H) = 1 - \hat{S}_i(H). \quad (5)$$

The resulting $\hat{p}_i(H)$ was scored against the same fixed-horizon binary labels as the classifiers.

B. Calibration and shared-cohort evaluation

Tree finalists were evaluated with raw probabilities and with Platt [14] or isotonic [15] calibration fitted only on the calibration subset, preserving the held-out test subset for final evaluation. Because model families can produce slightly different evaluable test rows, final PR-AUC, ROC-AUC, and Brier values were recomputed on the shared row intersection

$$\mathcal{D}_h^{\text{shared}} = \bigcap_k \mathcal{D}_h^{(k)} \quad (6)$$

where $\mathcal{D}_h^{(k)}$ denotes the evaluable row set for model family k at horizon h , so every retained model was scored on identical patients, years, and targets.

C. Intervention-safe track

The second track imposed monotonic constraints so that favorable feature changes could not raise predicted risk. FBS, cumulative maximum FBS, BMI, waist, and sugary-drink intake were constrained to increase risk, while exercise, physical activity, and fruit/vegetable intake were constrained to be protective. Monotonic XGBoost, LightGBM, CatBoost, Explainable Boosting Machines (EBM) [23], and logistic regression were benchmarked. Each constrained feature x_j carried a sign $c_j \in \{-1, 0, 1\}$ that forced the model to be non-increasing, unconstrained, or non-decreasing in x_j . The predicted probability was mapped to a horizon-specific score

$S = 100\hat{p}$. For each favorable scenario m (lower sugar, more exercise, lower BMI, etc.), modified values were clipped to the observed training range and the score change was $\Delta S_i^{(m)} = S_{i,\text{int}}^{(m)} - S_{i,\text{base}}$. A model passed only if no patient’s score rose beyond a small tolerance ϵ :

$$\text{PASS} \iff \forall i \in \mathcal{D}_{\text{test}}, \forall m \in \mathcal{M}_{\text{fav}}, \Delta S_i^{(m)} \leq \epsilon. \quad (7)$$

D. Metrics

The primary metric was precision-recall AUC (PR-AUC), appropriate because the positive class was a minority [31]. Its random baseline equaled the positive prevalence, which grew with N , so PR-AUC was read against the per-horizon baseline (lift over random). ROC-AUC (fixed 0.5 baseline) and the Brier score supported discrimination and calibration assessment.

V. RESULTS

A. Pure prediction

The shared-cohort leaderboard contained 28 model instances per horizon (140 in total) spanning tree, statistical, and survival approaches; the top five per horizon appeared in Table I. Two patterns stood out. First, the winning family changed with horizon: CatBoost led at one year, Logistic narrowly led at two years, XGBoost led at three years, Logistic won at four years, and GEE won at five years. Trees won two of the first three horizons and occupied most top-ranked rows from $N=1$ to $N=3$, while longer horizons shifted toward statistical models. Second, the families were close: the rank-1 to rank-5 PR-AUC gap was at most 0.014 at every horizon, so the transition from nonlinear learners to statistical models was gradual rather than abrupt. To test whether these narrow gaps were meaningful, we computed patient-clustered bootstrap 95% confidence intervals ($B=2000$, resampling patients rather than rows to respect within-patient correlation) [32]. At every horizon the difference between the best tree and the best biostatistical model had a 95% CI that spanned zero. The per-horizon leaders were therefore statistically indistinguishable. All winners sat at ROC-AUC 0.80 to 0.83, within the conventional “excellent discrimination” band, even though absolute PR-AUC increased with N because prevalence rose.

B. Ablations: history window, survival framing, and calendar-time features

We ran three ablations. First, on the history window: across all horizons, $M=5$ gave the strongest unconstrained tree, and the calibrated tree finalists were likewise drawn from $M=5$; ROC-AUC decreased gently from about 0.83 at $N=1$ to 0.80 at $N=5$, as expected for longer-range forecasting. Second, on survival framing: two-stage prediction beat landmark Cox at every horizon (by ~ 0.01 – 0.02 PR-AUC), but a one-year history collapsed it to a degenerate ranker at the event prevalence. This confirmed that temporal framing was part of the problem definition, not a tuning detail. Even at its best settings the survival family never topped the leaderboard: it was statistically comparable to the per-horizon leader at

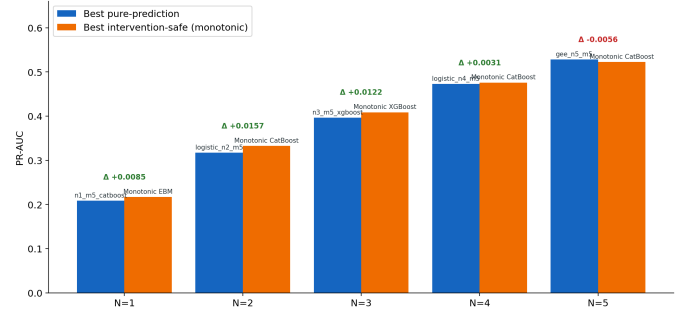


Fig. 2. Best intervention-safe model versus best pure predictor per horizon. The intervention-safe model was at or above the pure-prediction winner at horizons one to four (positive Δ) and trailed only marginally at five years, while preserving directional safety.

one to three years but significantly behind at four and five (bootstrap CI on the gap excluding zero), so it served as a comparator rather than the final recommendation. Third, on calendar-time features: the three engineered calendar terms (the observation year and its centered and squared forms) were properties of the cohort, not the patient. Removing them cost a mean 0.009 PR-AUC, concentrated at the one-year horizon (-0.021) and negligible beyond $N=3$ (within the ± 0.002 run-to-run noise). The calendar signal therefore aided only near-term forecasting [29].

C. Ablation: feature engineering

The enriched feature set added clinically motivated nonlinear and interaction terms (FBS hinged at the 100 and 125 mg/dL thresholds, $\text{FBS} \times \text{Age}$ and $\text{MAX_FBS} \times \text{Age}$ interactions, and a quadratic time trend). On the shared one-year test setting both statistical families improved on every metric, with logistic regression gaining the most (PR-AUC $0.174 \rightarrow 0.191$, ROC-AUC $0.808 \rightarrow 0.816$; GEE $0.175 \rightarrow 0.186$). The gain was specific to the statistical family: the trees were essentially unchanged, because a boosted learner could already recover a threshold hinge or a multiplicative interaction through successive splits. All final models nonetheless used the enriched set so that the same clinical domain knowledge was represented identically across families.

D. Intervention-safe benchmark

On identical shared-cohort test rows, the intervention-safe winners matched or exceeded the pure-prediction winner at horizons one to four and trailed only marginally at five years (Fig. 2), while every benchmarked monotonic family was perfectly directionally safe (directionally-correct rate 1.0, zero unexpected increases under the tested presets). Monotonic CatBoost was the strongest single family, winning at two, four, and five years.

E. Feature importance and interpretability

Because the leaderboard spanned several model families, we characterized importance at two levels. At the *data level*, model-agnostic univariate effect sizes were largest for fasting glucose and its cumulative maximum (Cohen’s $d \approx 1.02$ each),

TABLE I

TOP-5 PURE-PREDICTION RESULTS PER HORIZON ON THE SHARED-COHORT LEADERBOARD. EACH HORIZON HEADER GIVES THE POSITIVE PREVALENCE, THE RANDOM-CLASSIFIER PR-AUC BASELINE (EQUAL TO THAT PREVALENCE), AND THE RANK-1 LIFT OVER RANDOM. TREE MODELS DOMINATED OR REMAINED NEAR THE TOP AT SHORT HORIZONS, WHILE STATISTICAL MODELS LED AT LONGER HORIZONS.

Rank	Approach	Family	M	Calib.	PR-AUC	ROC-AUC	Brier
<i>Horizon $N=1$: positive rate 3.86%, random PR-AUC 0.0386, rank-1 lift $5.41\times$</i>							
1	Tree	CatBoost	5	Raw	0.2085	0.8297	0.0335
2	Tree	CatBoost	5	Platt	0.2085	0.8297	0.0335
3	Tree	XGBoost	5	Platt	0.2083	0.8272	0.0335
4	Tree	XGBoost	5	Raw	0.2083	0.8272	0.0335
5	Survival	Two-stage	5	Raw	0.2005	0.8176	0.0340
<i>Horizon $N=2$: positive rate 7.07%, random PR-AUC 0.0707, rank-1 lift $4.48\times$</i>							
1	Statistical	Logistic	5	Raw	0.3169	0.8228	0.0562
2	Tree	XGBoost	5	Raw	0.3160	0.8253	0.0561
3	Tree	XGBoost	5	Platt	0.3160	0.8253	0.0562
4	Tree	CatBoost	5	Platt	0.3155	0.8271	0.0562
5	Tree	CatBoost	5	Raw	0.3155	0.8271	0.0562
<i>Horizon $N=3$: positive rate 10.27%, random PR-AUC 0.1027, rank-1 lift $3.86\times$</i>							
1	Tree	XGBoost	5	Platt	0.3964	0.8157	0.0761
2	Tree	XGBoost	5	Raw	0.3964	0.8157	0.0760
3	Tree	CatBoost	5	Raw	0.3953	0.8187	0.0759
4	Tree	CatBoost	5	Platt	0.3953	0.8187	0.0759
5	Statistical	Logistic	5	Raw	0.3926	0.8171	0.0757
<i>Horizon $N=4$: positive rate 13.46%, random PR-AUC 0.1346, rank-1 lift $3.51\times$</i>							
1	Statistical	Logistic	5	Raw	0.4727	0.8097	0.0921
2	Statistical	Logistic	3	Raw	0.4651	0.8100	0.0925
3	Statistical	GEE	5	Raw	0.4639	0.8064	0.0939
4	Tree	CatBoost	5	Platt	0.4616	0.8093	0.0928
5	Tree	CatBoost	5	Raw	0.4616	0.8093	0.0927
<i>Horizon $N=5$: positive rate 16.49%, random PR-AUC 0.1649, rank-1 lift $3.20\times$</i>							
1	Statistical	GEE	5	Raw	0.5282	0.8048	0.1075
2	Statistical	Logistic	5	Raw	0.5248	0.8051	0.1067
3	Statistical	Logistic	3	Raw	0.5190	0.8046	0.1071
4	Tree	CatBoost	5	Platt	0.5150	0.8052	0.1075
5	Tree	CatBoost	5	Raw	0.5150	0.8052	0.1074

more than double the next strongest predictor, waist circumference ($d \approx 0.55$). At the *model level*, the interpretable statistical winner agreed: the GEE coefficient forest for the five-year model placed the rolling five-year mean and cumulative-maximum FBS as the largest positive, robustly significant associations ($p < 0.05$ under cluster-robust variance) [29]. Extending this across families, the rolling five-year mean FBS was the most important feature overall and ranked among the top features of every per-horizon winning model (CatBoost, XGBoost, GEE, and logistic). The dominant glycemic signal was therefore consistent across the winning statistical and tree families, not an artifact of one [29].

VI. DISCUSSION

The results argued against a universal “best model”: the nominal best family shifted with the forecast horizon, with trees leading at short horizons and repeated-measures statistical models at longer ones, though the two were statistically indistinguishable within bootstrap confidence intervals. The survival branch clarified the target framing: classification-aligned formulations matched fixed-horizon binary onset more directly, although two-stage survival remained a strong comparator when adequate history was retained.

The dual-track view was the main practical message: passive screening and patient-facing what-if scoring shared the same data foundation and split but had different operational requirements, and conflating them risked deploying a model that behaved implausibly under favorable lifestyle changes. Notably, monotonic constraints imposed no accuracy cost: several intervention-safe models sat on or above the pure-prediction frontier, suggesting that clinically informed constraints could also regularize implausible patterns in observational data.

From a deployment standpoint, both tracks were lightweight: the final models were gradient-boosting and linear learners that ran on commodity CPU without specialized hardware, so per-patient risk scores could be produced with low latency in a standard clinical IT environment [29].

A. Limitations

The study had several limitations. The data were observational and from a single tertiary center, so external, multi-site validation is required before generalizability can be claimed. Follow-up was irregular and missingness informative; explicit indicators mitigated but did not eliminate the resulting bias, and the shared-cohort evaluation was weighted toward patient-years with more complete measurements, so full-population deployment needs monitoring under sparser data.

The intervention-safe scores were directionally constrained but were not causal estimates of benefit, since observational lifestyle changes are confounded and mutually correlated. Finally, calibration may drift as the care environment shifts, so periodic recalibration or retraining is needed for models served beyond the training window.

VII. CONCLUSION

We presented a longitudinal diabetes-risk pipeline on twelve years of health-check data that treated prediction as a dual-track problem. No single family won across horizons, longer history helped the direct-prediction finalists, and intervention-safe monotonic models preserved clinically plausible score direction while matching or exceeding the pure-prediction winners at four of five horizons, yielding two complementary tools, a screening leaderboard and an intervention-safe benchmark, on a common foundation. Future work centers on external, multi-site validation to establish generalizability beyond a single center. Further directions include calibration monitoring with periodic retraining against temporal drift, causal estimation of intervention effects to move the intervention-safe track beyond directional safety, and recurrent or Transformer-based sequence models [25], [26] as an additional comparator once longer, denser records accumulate. Code was released for reproducibility [29].

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